

MATH 2 FINAL COACHING

81 items

QUESTIONS	SOLUTION:
1. The processing load of a system is modeled by the function $E(x, y) = x^2 + 4xy + y^2$ where x represents the processing load and y represents another operating parameter. Find the partial derivative of E with respect to the processing load (x). A. $2x + 4y$ B. $2x + 2y$ C. $4x + y$ D. $x + 4y$	$E(x, y) = x^2 + 4xy + y^2$ $\frac{\partial E}{\partial x} = 2x + 4y$
2. A scalar field is given by $F(x, y) = x^2 + y^2$ Find the $\nabla \cdot F$. A. $x+y$ B. $2x+2y$ C. $2x+y$ D. x^2+y^2	$F(x, y) = x^2 + y^2$ $\nabla \cdot F = \frac{dfx}{dx} + \frac{dfy}{dy}$ $\nabla \cdot F = 2x + 2y$
3. The cost function of a company is given by $C(x, y) = 5x^2 + 3xy + 2y^2 + 10$ where x and y represent the quantities of two products. What is the marginal cost with respect to x when x = 2 and y = 3? A. 23 B. 26 C. 29 D. 32	$C(x, y) = 5x^2 + 3xy + 2y^2 + 10$ $\text{Calc: } \frac{d}{dx} (5x^2 + 3xy + 2y^2 + 10) \Big _{x=2}$ $\text{Calc: } x = 2$ $y = 3$ ANS. 29
4. A diode current is given by	

$$I(V, T) = I_0 (e^{(qV/kT)} - 1)$$

Differentiate the equation with respect to temperature T assuming V is constant. Which of the following represents dI/dT ?

- A. $-(qV/kT^2) I_0 e^{(qV/kT)} + (dI_0/dT)(e^{(qV/kT)} - 1)$
- B. $(qV/kT^2) I_0 e^{(qV/kT)} + (dI_0/dT)(e^{(qV/kT)} - 1)$
- C. $(qV/kT) I_0 e^{(qV/kT)}$
- D. $(dI_0/dT)e^{(qV/kT)}$

5. 5. A curve is defined parametrically by $x = t^2 + 1$ and $y = t^3 - 2$. Find dy/dx when $t = 1$.

- A. 1
- B. 3/2
- C. 2/3
- D. 3

Parametric Curve:

Calc:
$$\frac{dy}{dx} = \frac{\frac{d}{dt}(\text{Given } y)}{\frac{d}{dt}(\text{Given } x)} \bigg|_{t=\text{Given}}$$

Given:

$$\begin{aligned} x &= t^2 + 1 \\ y &= t^3 - 2 \\ t &= 1 \end{aligned} \quad \frac{dy}{dx} = \frac{\frac{d}{dt}(t^3 - 2)}{\frac{d}{dt}(t^2 + 1)} \bigg|_{t=1}$$

Ans. $\frac{3}{2}$

6. The position of a particle is given by the parametric equations: $x = 3 \cos(t)$ and $y = 3 \sin(t)$

Find the **speed of the particle**.

- A. 1
- B. 2
- C. 3
- D. 6

Speed of Particle (Parametric)

$$v(t) = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

$$\frac{dx}{dt} = -3 \sin(t) \quad v(t) = \sqrt{(-3 \sin(t))^2 + (3 \cos(t))^2}$$

$$\frac{dy}{dt} = 3 \cos(t)$$

$v(t) = 3$

7. A particle moves in the xy-plane such that its position at time t is given by the parametric equations:

- $x = t^2 + t$ and $y = 3t - 1$. What are the coordinates of the particle at the initial time $t = 0$?

- A. (0, -1)
B. (1, -1)
C. (0, 1)
D. (1, 0)

@ $t = 0$

$$x = 0^2 + 0 = 0$$

$$y = 3(0) - 1 = -1$$

ANS. (0, -1)

8. Find the equation of the tangent line to the curve defined by the parametric equations: $x = t^2 + t$ and $y = 3t - 2$ at $t = 2$.

- A. $y = (3/5)x - 2/5$
B. $y = (3/5)x + 2/5$
C. $y = (5/3)x - 2/5$
D. $y = (3/2)x + 1$

$$\frac{dx}{dy} = \frac{\frac{d}{dt}(t^2 + t)}{\frac{d}{dt}(3t - 2)} \bigg|_{t=2} = \frac{3}{5}$$

Tangent line:

$$y - y_1 = m(x - x_1)$$

$$y - 4 = \frac{3}{5}(x - 6)$$

$$y = \frac{3}{5}(x - 6) + 4$$

$$y = \frac{3}{5}x + \frac{2}{5}$$

$$x(2) = 2^2 + 2 = 6$$

$$y(2) = 3(2) - 2 = 4$$

points (6, 4)

9. In an oscilloscope measurement, a voltage signal

$v(t) = 10 \sin(\omega t)$ is applied to the horizontal input, and a current signal $i(t) = 5 \cos(\omega t)$ is applied to the vertical input. What geometric figure is produced in the V-I display?

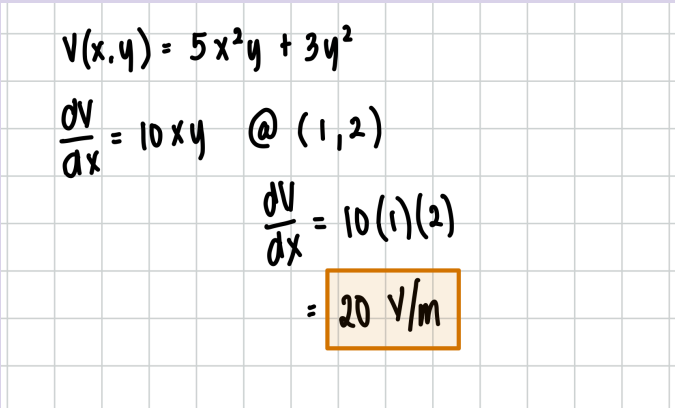
- A. Straight line
B. Circle
C. **Ellipse**
D. Parabola

10. For a semiconductor device, the electric potential is given by

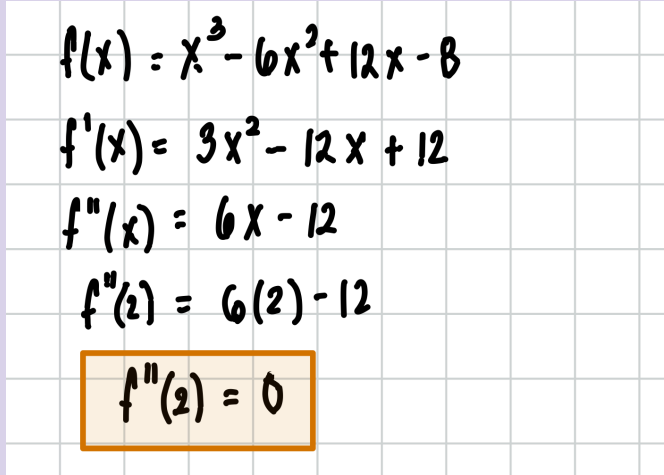
$$V(x, y) = Ax^2 + By^2$$

For the potential to satisfy Laplace's Equation, what must be the relationship between A and B?

- A. $A = B$
B. **$A = -B$**
C. $A = 2B$
D. $B = 0$

11. An electric potential field is given by $V(x, y) = 3x^2 + 2y^2$ A particle with charge $2 \mu\text{C}$ is placed at the point $(2,1)$. What is the force acting on the particle? A. (12, 16) μN B. (16, 12) μN C. (24, 8) μN D. (8, 24) μN	
12. In a certain region of space, the electric potential is given by $V(x, y) = 5x^2y + 3y^2$ where V is in volts and x and y are in meters. Find the rate of change of the potential with respect to x at the point $(1,2)$. A. 10 V/m B. 20 V/m C. 24 V/m D. 30 V/m	 $V(x,y) = 5x^2y + 3y^2$ $\frac{dV}{dx} = 10xy \quad @ (1,2)$ $\frac{dV}{dx} = 10(1)(2)$ $= 20 \text{ V/m}$
13. Given the function $f(x) = x^3 - 6x^2 + 9x + 7$ on the closed interval $[0,4]$, find the absolute maximum value of the function. A. 4 B. 5 C. 7 D. 8 *DI BA 11? "ou haha"	
14. What is the absolute minimum value of $f(x) = x^3 - 12x + 7$ on the interval $[-3,3]$? A. -3 B. -2 C. 2	

D. 3	
15. An electronic engineer is testing the output of a signal generator. the waveform is expressed in polar form $10 \angle (2000t + 45)$. This representation comes from modeling the signal as a sum of two phasors rotating in opposite directions. Which of the following is the equivalent real time-domain signal? A. $10\cos(2000t + 45)$ B. $20\sin(2000t + 45)$ C. $20\cos(2000t + 45)$ D. $10\sin(2000t + 45)$	
16. An electric motor winding is modelled as a series RL circuit where $R=4$, $L=2$. At $t=0$, a constant voltage $V=20V$ is applied. Which of the following expresses current as a function of time? A. $5(1 - e^{-2t})$ B. $10(1 - e^{-2t})$ C. $5e^{-2t}$ D. $20(1 - e^{-2t})$	(Increasing) $i(t) = \frac{V}{R} (1 - e^{-R/L t})$ $i(t) = \frac{20}{4} (1 - e^{-4/2 t})$ $i(t) = 5(1 - e^{-2t})$
17. The electric potential is given $V(x,y) = 4x^2y + y^3$. What is the magnitude of the electric field at point (1,1) A. $\sqrt{97}$ B. $\sqrt{113}$ C. $\sqrt{121}$ D. $\sqrt{145}$	
18. A delivery drone launches a small package from a roof top at a height of $h=10m$ from the ground. The package is released with an initial velocity of $15m/s$ at an angle of 60 degrees from the ground. The parametric equations that describe its motion are: $x(t) = V_0 \cos t$ $y(t) = h + V_0 \sin t - \frac{1}{2} g t^2$ given that $g=9.8 m/s^2$. How long until the package reaches the ground?	

A. 2.74 s B. 3.27 s C. 3.81 s D. 4.10 s	
19. A ball is launched with an initial velocity of 20m/s at an angle of 45 degrees from the ground. The parametric equations that describe its motion are: $x(t) = 20\cos 45t$ and $y(t) = 20\sin 45t$. How long until the ball reaches its maximum height? A. 1.02 s B. 1.44 s C. 2.04 s D. 2.88 s	
20. Given $f(x) = x^3 - 6x^2 + 12x - 8$. What is its value at its inflection point? A. -8 B. -2 C. 0 D. 2	
21. Given $f(x) = x^3 - 6x^2 + 12x - 8$. What is its concavity? A. Concave up for $x < 2$, concave down for $x > 2$ B. Concave up for all x C. Concave down for $x < 2$, concave up for $x > 2$ D. No concavity change	
22. For the function $f(x) = x^3 - 6x^2 + 12x - 8$. Find the second derivative at $x = 2$. A. -12 B. -6 C. 0 D. 6	 $f(x) = x^3 - 6x^2 + 12x - 8$ $f'(x) = 3x^2 - 12x + 12$ $f''(x) = 6x - 12$ $f''(2) = 6(2) - 12$ $f''(2) = 0$

23. Find the slope of the tangent line to the curve $x \ln(y) - 3x + y = 0$ at the point (2,1). A. -2 B. -1 C. 1 D. 2	
24. Find the slope of the normal line to the curve $x \ln(y) - 3x + y = 0$ at the point (2,1). A. -2 B. -1 C. 1 D. 2	
25. Evaluate the limit $\lim_{(x \rightarrow 0)} 2x / (7x)$ A. 2/7 B. -2/7 C. 0 D. DNE	
26. Evaluate the limit $\lim_{(t \rightarrow 2)} 3t^2 / (5t + 1)^3$ A. 12/121 B. 12/1331 C. 6/121 D. 4/1331	
27. Evaluate the limit $\lim_{(x \rightarrow 1)} (\sqrt{x} - 1) / (x - 1)$ A. 1/4 B. 1/3 C. 1/2 D. 1	

28. $\lim_{x \rightarrow 0} (e^{2x} - 1) / x$ A. 1 B. 2 C. e D. 2e	
29. $\lim_{x \rightarrow 0^+} x \ln(x)$ A. $-\infty$ B. -1 C. 0 D. DNE	
30. The power dissipated in a circuit depends on the currents I_1 and I_2 according to $P(I_1, I_2) = 10I_1 + 15I_2 - I_1^2 - 2I_2^2 - I_1I_2$ where P is in watts and I_1 and I_2 are in amperes. What values of I_1 and I_2 maximize the power? A. $I_1 = 3.57$ A, $I_2 = 2.86$ A B. $I_1 = 1.54$ A, $I_2 = 2.43$ A C. $I_1 = 2.5$ A, $I_2 = 3.68$ A D. $I_1 = 4$ A, $I_2 = 2$ A	
31. Evaluate the double integral $M = \iint_R (x + y) \, dA$ where R is the rectangle defined by $0 \leq x \leq 1, 0 \leq y \leq 2$ A. 2 B. 3 C. 4 D. 5	
32. A thin plate occupies the region $0 < x < 2$, $0 < y < x$ with uniform density. Find the x-coordinate of the centroid.	

A. $1/2$ B. $2/3$ C. 1 D. $4/3$	
33. What is the average value of $\sin x$ over the following intervals $(0, \pi)$, $(-\pi, \pi)$, $(0, 2\pi)$? A. $(0.6366, 0, 0)$ B. $(0, 0.6366, 0)$ C. $(0.6366, 0.6366, 0)$ D. $(0, 0, 0.6366)$	
34. What is the average value of $\cos x$ over the following intervals $(0, \pi/2)$, $(-\pi/2, \pi/2)$, $(0, \pi)$? A. $(0.6366, 0.6366, 0)$ B. $(0, 0.6366, 0.6366)$ C. $(0.6366, 0, 0)$ D. $(0, 0, 0)$	
35. If the covariance between two variables X and Y is given as $\text{cov}(X, Y) = 1$ and their variances are $\text{var}(X) = 1$, $\text{var}(Y) = 4$ what is the correlation coefficient? A. 0.25 B. 0.5 C. 1 D. 2	
36. If the correlation coefficient between two variables X and Y is 0.5 and their variances are $\text{var}(X) = 1$ and $\text{var}(Y) = 4$, what is the covariance $\text{cov}(X, Y)$? A. 0.25 B. 0.5 C. 1 D. 2	

37. A researcher studies the relationship between hours of study (X) and exam scores (Y). The following statistics are obtained: $\text{cov}(X,Y) = 6$, $\text{var}(X) = 9$, $\text{var}(Y) = 16$. What is the correlation coefficient? A. 0.25 B. 0.50 C. 0.75 D. 1.00	
38. A 20 μF capacitor is initially charged to 50 V. It is disconnected from the source and allowed to discharge through a 50 kΩ resistor. After 1 second, what is the capacitor voltage? A. 3.52 V B. 5.52 V C. 8.21 V D. 10.50 V	
39. An emergency LED system requires the capacitor voltage to remain above 9 V for at least 5 seconds after power loss. The capacitor is initially charged to 15 V and discharges through a 30 kΩ load resistance. What is the minimum required capacitance? A. 220 μF B. 270 μF C. 326 μF D. 470 μF	
40. What is the first-order differential equation that represents the current $i(t)$ in an RL circuit with resistance R and inductance L? A. $L(di/dt) + Ri = V(t)$ B. $L(di/dt) - Ri = 0$ C. $L(di/dt) + Ri = 0$ D. $L(di/dt) + Li = V(t)$	

41. A series RL circuit with resistance R and inductor L is carrying an initial current I_0 . At $t = 0$, the voltage source is disconnected. Which first-order differential equation correctly describes the current for $t > 0$?

- A. $L(di/dt) + Ri = V(t)$
- B. $L(di/dt) - Ri = 0$
- C. $L(di/dt) + Ri = 0$**
- D. $L(di/dt) = Ri$

42. A series RL circuit with resistance R and inductor L is driven by a DC source V. The circuit has reached steady state. At $t = 0$, the source suddenly reverses polarity while R and L remain unchanged. Which first-order differential equation correctly governs the current for $t > 0$?

- A. $L(di/dt) + Ri = V$
- B. $L(di/dt) + Ri = -V$**
- C. $L(di/dt) - Ri = V$
- D. $R(di/dt) - Li = -V$

43. A series RL circuit has $R = 10 \text{ ohms}$ and $L = 2 \text{ H}$. The applied voltage source is a step function:

$$V(t) = 0 \text{ for } t < 0$$

$$V(t) = 20 \text{ for } t > 0$$

The initial current is zero. Which piecewise differential equation correctly describes the current for $t > 0$?

- A. $2(di/dt) + 10i = 20, i(0) = 0$**
- B. $10(di/dt) + 2i = 20, i(0) = 0$
- C. $2(di/dt) + 10i = 0, i(0) = 20$
- D. $2(di/dt) - 10i = 20, i(0) = 0$

$$R = 10 \Omega \quad \left\{ \begin{array}{ll} 0 & t < 0 \\ 20 & t > 0 \end{array} \right\}$$

$$L = 2 \text{ H}$$

$$L \frac{di}{dt} + Ri = V(t)$$

$$2 \frac{di}{dt} + 10i = 20, i(0) = 0$$

44. Solve for the $I(t)$ of the above problem.

A. $I(t) = 2(1 - e^{-5t})$ A

B. $I(t) = 2e^{-5t}$ A

C. $I(t) = 20(1 - e^{-5t})$ A

D. $I(t) = 2(1 - e^{-t/5})$ A

(Increasing)

$$i(t) = \frac{V}{R} (1 - e^{-R/L t})$$

$$i(t) = \frac{20}{10} (1 - e^{-R/L t})$$

$$i(t) = 2(1 - e^{-5t})$$

45. What is the length of the curve $y = x^3/3$ from $x = 0$ to $x = 2$?

A. $\int_0^2 \sqrt{1 + x^2} dx$

B. $\int_0^2 \sqrt{1 + x^4} dx$

C. $\int_0^2 \sqrt{1 + x^3} dx$

D. $\int_0^2 \sqrt{1 + 3x^2} dx$

Length of Curve:

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$L = \int_0^2 \sqrt{1 + \left(\frac{1}{3}(3x^2)\right)^2} dx$$

$$L = \int_0^2 \sqrt{1 + x^4} dx$$

46. For the curve $y = x^3/3$, $0 \leq x \leq 2$, which integral represents the surface area generated when the curve is revolved about the x-axis?

A. $\int_0^2 2\pi x \sqrt{1 + x^4} dx$

B. $\int_0^2 (2\pi x^3/3) \sqrt{1 + x^4} dx$

C. $\int_0^2 (2\pi x^2/3) \sqrt{1 + x^4} dx$

D. $\int_0^2 2\pi x^2 \sqrt{1 + x^4} dx$

47. A curve $y = \sin(x)$, $0 \leq x \leq \pi$, is rotated about the x-axis. Which integral represents the volume?

A. $\int_0^\pi \pi \sin(x) dx$

B. $\int_0^\pi \pi \sin^2(x) dx$

C. $\int_0^\pi 2\pi \sin(x) dx$

D. $\int_0^\pi \pi \cos^2(x) dx$	
48. A spring has a position dependent stiffness $k(x)$, so the $F = k(x) dx$. Work is done stretching the spring from $x=0$ to $x=X$. What represents the work? A. $\int_0^x k(x) dx$ B. $\int_0^x k(x)x dx$ C. $\int_0^x x^2 dx$ D. $\int_0^x k(x)^2 dx$	
49. A vertical wall has a height h and width b is submerged to a liquid of density p. Pressure increases with depth y (from the surface) so $P(y) = pgy$. Which integral represents the hydrostatic Force? A. $\int_0^h pgy dy$ B. $\int_0^h pgyb dy$ C. $\int_0^h pg y^2 b dy$ D. $\int_0^h pgb dy$	
50. A thin rod of length L and mass M lies along the x-axis. Find the moment of inertia about one end. A. $\int_0^L (M/L) dx$ B. $\int_0^L x (M/L) dx$ C. $\int_0^L x^2 (M/L) dx$ D. $\int_0^L x M dx$	
51. A thin rod of length L lies along the x-axis from $x = 0$ to $x = L$ with uniform mass distribution. Which expression gives the x-coordinate of the centroid? A. $(\int_0^L x^2 dx) / (\int_0^L dx)$ B. $(\int_0^L x dx) / (\int_0^L dx)$ C. $\int_0^L x dx$ D. $(\int_0^L dx) / (\int_0^L x dx)$	

52. What is the solution of the first-order differential equation $dy/dx = x/y$?

- A. $y^2 = x^2 + C$
- B. $y = x + C$
- C. $y^2 = x + C$
- D. $y = x^2 + C$

$$\begin{aligned}\frac{dy}{dx} &= \frac{x}{y} \\ \int y dy &= \int x dx \\ \frac{y^2}{2} &= \frac{x^2}{2} + C \\ &= y^2 = x^2 + C\end{aligned}$$

53. For the sequence $x[n] = (1, 2, 1)$, what is the autocorrelation at lag 0?

- A. 3
- B. 4
- C. 5
- D. 6

$$\begin{aligned}R_{xx}[0] &= \sum_n (x[n])^2 \\ x[n] &= 1^2 + 2^2 + 1^2 \\ &= 6\end{aligned}$$

54. Given two sequences $x[n] = (1, 2)$ and $y[n] = (2, 1)$, what is the cross-correlation at lag 0?

- A. 2
- B. 3
- C. 4
- D. 5

$$\begin{aligned}r_{xy}[0] &= \sum_n x[n] \cdot y[n] \\ (1, 2) &(2, 1) \\ r_{xy} &= 2 + 2 = 4\end{aligned}$$

55. If $x[n]$ is a real-valued discrete-time sequence, which property of the autocorrelation function is true?

- A. $R_{xx}[k] = -R_{xx}[-k]$
B. $R_{xx}[k] = R_{xx}[-k]$
 C. $R_{xx}[k] = 0$
 D. $R_{xx}[k] = k$

56. What is the linear convolution of the two discrete signals $x[n] = (1, 2)$ and $h[n] = (3, 4)$?

- A. (3, 8, 8)
B. (3, 10, 8)
 C. (4, 10, 6)
 D. (3, 6, 8)

$x[n] \downarrow$ $h[n] \rightarrow$ 3 4
 1 $(1)(3) = 3$ $(1)(4) = 4$
 2 $(2)(3) = 6$ $(2)(4) = 8$
 $y[0] = 3$ $L = 2 + 2 - 1 = 3$ $y[0], y[1], y[2]$
 $y[1] = 6 + 4 = 10$
 $y[2] = 8$
ANS. (3, 10, 8)

57. If $\text{cov}(X, Y) = 1$, the variances are $\text{var}(X) = 1$ and $\text{var}(Y) = 4$. What is the correlation coefficient?

- A. 1/4
B. 1/2
 C. 1
 D. 2

$$\frac{\text{cov}(X, Y)}{\sqrt{\text{var}(x) \cdot \text{var}(y)}} = \frac{1}{\sqrt{(1)(4)}}$$
 $= \frac{1}{2}$

58. What is the **linear convolution** between two signals

$x[n] = (1, 2)$ and $h[n] = (3, 4)$?

- A. (3, 8, 8)**
 B. (3, 10, 8)
 C. (3, 6, 8)
 D. (1, 10, 4)

59. The power consumption of a machine is modeled by $P(t) = 3t^2 + 2t + 1$ W. Using the trapezoidal rule with $n = 1$, what is the

approximate total energy consumed from $t = 0$ to $t = 2$?

- A. 16 J
- B. 18 J**
- C. 20 J
- D. 22 J

60. A car's speed (m/s) at equal time intervals of 2 s is recorded as follows:

0	2	4	6
0	10	18	20

Using the **Trapezoidal Rule**, estimate the total distance traveled from $t = 0$ to $t = 6$.

- A. 72 m
- B. 74 m
- C. 76 m**
- D. 78 m

Trapezoidal: $h = 2s$

$$= \frac{h}{2} (f_0 + f_3 + 2(f_1 + f_2))$$

$$= \frac{2}{2} (0 + 20 + 2(10 + 18))$$

$$= 76 \text{ m}$$

Question 21

Evaluate the integral using five function values, using Trapezoidal Rule.

$f(x) = x \log(x)$

$h = \frac{8-1}{4} = \frac{7}{4} = 1.75$

Soln. Trapezoidal Rule

$A = A + \frac{h}{2} (f(x_0) + f(x_4))$

$A = 0 + \frac{1.75}{2} (0 + 8)$

$A = 7$

Answer: C. 6.7

61. The flow rate of the water in a pipe (m^3/s) is measured every 2 minutes.

0	2	4	6
3	5	4	6

Estimate the total volume of water that flowed in the pipe using Simpson's 1/3 rule.

- A. 24.67 m^3**
- B. 25.33 m^3
- C. 26.67 m^3
- D. 27.33 m^3

Simpson's 1/3 Rule: $h = 2 \text{ mins}$

$$= \frac{h}{3} (f_0 + f_3 + 4(f_1) + 2(f_2))$$

$$= \frac{2}{3} (3 + 6 + 4(5) + 2(4))$$

$$= 24.67 \text{ m}^3$$

Question 23

Approximate the following integral using Simpson's rule with $n = 4$.

$\int_{0.1}^{0.5} \frac{\cos x}{x} dx$

Soln.

$h = \frac{0.5 - 0.1}{4} = \frac{0.4}{4} = 0.1$

$A = A + \frac{h}{3} (f(x_0) + 4f(x_1) + f(x_2))$

$A = 0 + \frac{0.1}{3} (\frac{\cos(0.1)}{0.1} + 4 \frac{\cos(0.2)}{0.2} + \frac{\cos(0.3)}{0.3})$

$A = 1.5629$

Answer: B. 1.5629

62. Given the supply equation $P + Q = 10$ and the demand equation $P^2 + Q^2 = 58$, determine the possible equilibrium points (P, Q) .

- A. (3,7) and (7,3)
- B. (4,6) and (6,4)
- C. (5,5) and (3,7)
- D. (2,8) and (8,2)

63. Given the ODE: $L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{1}{C} i(t) = 0$, with $L = 1$, $R = 4$, $C = 0.25$, and initial conditions $i(0) = 0$ and $di/dt(0) = 8$ A/s, determine the expression for $i(t)$.

- A. $8e^{-2t}$
- B. $8te^{-2t}$
- C. $4te^{-2t}$
- D. $8te^{2t}$

$$L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{1}{C} i(t) = 0$$

$$\frac{d^2 i}{dt^2} + 4 \frac{di}{dt} + \frac{1}{0.25} i = 0$$

$$m^2 + 4m + 4 = 0$$

$$m = -2$$

$$y = Ce^{-2x} + Cxe^{-2x}$$

Condition 1: $i(0) = 0$

$$y = \cancel{Ce^{-2x}} + Cxe^{-2x}$$

Condition 2: $\frac{di}{dt}(0) = 8$ A/s

$$y = 8xe^{-2x}$$

64. Find the surface area generated by revolving $y = \sqrt{x}$, from $x = 1$ to $x = 4$, about the x-axis.

- A. $2\pi \int_1^4 \sqrt{x} \sqrt{1 + 1/(4x)} dx$
- B. $\pi \int_1^4 \sqrt{x} \sqrt{1 + 1/(4x)} dx$
- C. $2\pi \int_1^4 x \sqrt{1 + 1/(4x)} dx$
- D. $\pi \int_1^4 x \sqrt{1 + 1/(4x)} dx$

65. Find the volume generated when $y = \sin x$, $0 \leq x \leq \pi$, is revolved about the x-axis.

A. $\int_0^\pi \sin x \, dx$ B. $\pi \int_0^\pi \sin x \, dx$ C. $\pi \int_0^\pi \sin^2 x \, dx$ D. $2\pi \int_0^\pi \sin x \, dx$	
66. The region bounded by $y = 4 - x^2$ and $y = 0$ is revolved about the x-axis. Which volume expression is correct? A. $\pi \int_{-2}^2 (4 - x^2) \, dx$ B. $\pi \int_{-2}^2 (4 - x^2)^2 \, dx$ C. $2\pi \int_{-2}^2 (4 - x^2) \, dx$ D. $\pi \int_0^2 (4 - x^2)^2 \, dx$	
67. The general solution for the voltage across the inductor in a critically damped RLC series circuit is of the form: A. $v(t) = Ae^{-at}$ B. $v(t) = (A + Bt)e^{-at}$ C. $v(t) = A \cos(\omega t) + B \sin(\omega t)$ D. $v(t) = Ae^{-at} + Be^{-at}$	
68. Which Taguchi S/N ratio corresponds to the larger-the-better characteristic? A. $-10 \log[(1/n) \sum y^2]$ B. $-10 \log[(1/n) \sum (1/y^2)]$ C. $10 \log[(1/n) \sum (1/y^2)]$ D. $10 \log[(1/n) \sum y^2]$	
69. A hot metal rod is left in a constant-temperature room. The rate at which it cools is proportional to the difference between its temperature and the room temperature. Which differential equation does this imply? A. $dT/dt = k(T - T_s)$ B. $dT/dt = -k(T - T_s)$ C. $dT/dt = k(T + T_s)$	

D. $dT/dt = -k(T + T_s)$	
70. What is the best method to evaluate $\int (2x + 3) / (x^2 + 3x + 2) dx$? A. Substitution B. Integration by Parts C. Partial Fraction D. Trigonometric Substitution	
71. What is the most efficient way to solve $\int e^x \sin(x) dx$? A. Substitution B. Tabular Integration C. Partial Fraction D. Power Reduction Formula	
72. What is the best way to evaluate $\int \sin^2(x) dx$? A. Substitution B. Partial Fraction C. Integration by Parts D. Power Reduction Formula	
73. The region bounded by $y = x^2$, the y-axis, and the line $x = 2$ is revolved about the x-axis. Which integral gives the volume of the solid generated? A. $\pi \int_0^2 x^2 dx$ B. $\pi \int_0^2 x^4 dx$ C. $2\pi \int_0^2 x^2 dx$ D. $\pi \int_0^2 x^3 dx$	
74. A survey of 400 customers found that 96 made a purchase. Construct the 95% confidence interval for the true conversion rate. A. (0.180, 0.300) B. (0.198, 0.282) C. (0.210, 0.270)	

D. (0.150, 0.330)	
75. A political analyst wants to estimate the proportion of voters who support a certain candidate. The analyst desires a margin of error of 3% at a 95% confidence level. Assuming maximum variability, what is the minimum sample size required? A. 904 B. 1203 C. 1068 D. 1209	
76. A battery's lifetime is normally distributed with a mean of 10 hours and a standard deviation of 1.5 hours. What proportion of batteries will last at least 12 hours? A. 0.9082 B. 0.0918 C. 0.1330 D. 0.5000	$z = \frac{x - \mu}{\sigma}$ $z = \frac{12 - 10}{1.5} = 1.33$ $z > 1 = 1 - 0.91$ $= 0.09$ $z < 1 \rightarrow 68\%$ $z = 1 \rightarrow 84\%$ $z > 1 \rightarrow 91\%$
77. A clinic receives an average of 3 patients per hour. Using the Poisson distribution, what is the probability that exactly 5 patients will arrive in one hour? A. 0.043 B. 0.082 C. 0.101 D. 0.241	$p = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$ $p = \frac{e^{-3} \cdot 3^5}{5!} = 0.101$
78. Evaluate $\lim_{x \rightarrow 0} (\tan x - x)/x^3$ as x approaches 0 A. 1/3 B. 1/2	

C. 1/6 D. 0	
79.	
80.	